VELSANET ARCHITECTURE

White Paper Collection — Structural Analysis Report

Prepared for: ITU-T Study Group 13 (Q1/13 — Future Networks / IMT-2030)
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19 White Papers | 9 Architectural Layers | 6 Proposed Work Items

1. Executive Summary

This report provides a structural analysis of the Velsanet Architecture White Paper Collection, comprising 19 technical white papers published between October 2025 and February 2026. The analysis is prepared to support the ITU-T Study Group 13 review process under the IMT-2030 (6G) standardization programme.

Velsanet proposes a fundamental architectural departure from conventional Internet and 5G infrastructure. Rather than optimizing existing packet-switched, address-centric designs, Velsanet defines a new network space from first principles — topology precedes addressing, intelligence is native to structure, and physical path isolation replaces logical multiplexing.

The 19 white papers form a coherent, layered body of work spanning nine architectural domains: Foundation, Physical Layer, Topology, Mathematics, AI Architecture, Cognition, Identity & Path Management, Governance, and Operations. Each domain is fully documented and internally consistent. The collection constitutes a complete architectural specification suitable for ITU-T study item consideration.

This report identifies six proposed Work Items for SG13 consideration and assesses the completeness and readiness of each architectural component for standardization engagement.

2. White Paper Inventory

The following table provides a complete inventory of all 19 white papers in the Velsanet Architecture Collection.

No.	White Paper Title	Ver.	Core Content	Domain
#01	**Velsanet Network White Paper**	v2.0	Network Architecture Overview — Topology-First, Intent-Centric design philosophy	Foundation
#02	**Wired–Wireless Convergence**	v1.0	Unified convergence of wired and wireless layers within polyhedral topology	Physical Layer
#03	**Multi-Optical-Core Transceiver (MOCT)**	v1.1	Hardware: physically isolated parallel optical cores, 8-channel per core, MEMS alignment	Physical Layer
#04	**Matrix Architecture**	v1.1	Multi-plane matrix switching; structural isolation and channel assignment	Physical Layer
#05	**Spatial Genesis**	v1.0	Pre-coordinate topological space; 5 polyhedral primitives (T4, C6, O8, D12, I20)	Topology
#06	**Multimodal-to-Cube Memory Architecture**	v1.0	Multimodal signal encoding into Cube memory units at the PAI layer	Cognition
#07	**Mathematical White Paper**	v1.1	Multi-layer graph theory; Q7 hypercube; super-adjacency matrix; formal proofs	Mathematics
#08	**AI White Paper**	v1.3	Cognitive E2E architecture; DIKW instantiation; Cube–PAI–AAI–AsAI loop	AI Architecture
#09	**Three-Layer AI System**	v1.0	Critique of AGI; structural necessity of PAI/AAI/AsAI separation	AI Architecture
#10	**Network AI White Paper**	v1.3	Network-native AI governance; Velsa hierarchy; topology-aware intelligence	AI Architecture
#11	**IVGF Conceptual Existence**	v1.0	International Velsanet Governance Framework; structural sovereignty model	Governance
#12	**Foundation White Paper**	v1.1	Structural critique of IP/Internet architecture; first-principles replacement	Foundation
#13	**Node Color White Paper**	v1.0	Color-coded node taxonomy; structural role differentiation across layers	Topology
#14	**City-level E2E Management Center**	v1.0	Urban-scale deployment model; city–region–national integration	Operations
#15	**Identity White Paper**	v1.0	Network-native identity; manufacturing-time provisioning; topology-bound authority	Security/Identity

No.	White Paper Title	Ver.	Core Content	Domain
#16	**Path Formation White Paper**	v1.0	Stateful path formation; structural matching; no packet routing	**Path Management**
#17	**Path Lifecycle White Paper**	v1.0	Path continuity, mobility handling, handover semantics, failure recovery	**Path Management**
#18	**Network AI & Global Governance**	v1.0	Projection-based governance; Q2–Q7 hierarchy; sovereign layer encoding	**Governance**
#19	**Cognition Cube System**	v1.0	SEU → 8-channel decomposition → Meta abstraction → Cube synthesis pipeline	**Cognition**

3. Architectural Layer Analysis

The 19 white papers can be grouped into nine architectural layers, each addressing a distinct design question. This layered structure demonstrates the systematic completeness of the Velsanet specification — from physical substrate to planetary governance.

Architectural Layer	White Papers	Design Question	Key Contribution	ITU-T Relevance
Layer 1 Foundation	**#01, #12**	Why Velsanet exists	Establishes the philosophical and structural critique of IP-based internet. Defines Topology-First as the foundational design principle.	SG13 Q1/13: Future Networks
Layer 2 Physical	**#02, #03, #04**	How it is built	Physically isolated parallel optical cores (MOCT), wired-wireless convergence, multi-plane matrix architecture. Eliminates packet-level non-determinism at hardware level.	SG13 Q6/13: Optical Transport
Layer 3 Topology	**#05, #13**	How space is defined	Pre-coordinate polyhedral space from 5 primitives (T4–I20). Complete and closed grammar for all network topologies. Node color taxonomy for role differentiation.	SG13 Q1/13: Architecture
Layer 4 Mathematics	**#07**	Formal proof basis	Multi-layer graph theory, Q7 hypercube mapping, super-adjacency matrix. Formal mathematical grounding for all structural claims.	SG13 Q1/13: Formal Models
Layer 5 AI Architecture	**#08, #09, #10**	How intelligence is embedded	PAI–AAI–AsAI three-layer cognitive architecture native to topology. DIKW instantiation as executable network structure. Replaces external AI overlay.	SG13 Q4/13: AI in Networks
Layer 6 Cognition	**#06, #19**	How cognition is generated	SEU → 8-channel decomposition → Meta abstraction → Cognition Cube pipeline. Structure-first cognition precedes model execution.	SG13 Q4/13: Edge Intelligence
Layer 7 Identity & Path	**#15, #16, #17**	How connections are formed	Manufacturing-time identity provisioning. Stateful path formation by structural matching. No packet routing. Identity-bound lifecycle.	SG13 Q7/13: Security
Layer 8 Governance	**#11, #18**	How sovereignty is preserved	Projection-based Q2–Q7 hierarchy. IVGF structural sovereignty model. Higher layers project states — do not issue commands.	SG13 Q3/13: Governance
Layer 9 Operations	**#14**	How cities are managed	City-scale E2E management; integration of urban, regional, and national layers. Operational deployment model.	SG13 Q5/13: Management

3.1 Structural Logic: Why the Layers are Ordered This Way

The layering is not arbitrary. Each layer logically depends on the one below it:

- Foundation (#01, #12) establishes why existing architecture must be replaced — not patched.

- Physical Layer (#02–04) defines the substrate on which topology can be realized.
- Topology (#05, #13) and Mathematics (#07) provide the formal space and proof basis.
- AI Architecture (#08–10) embeds intelligence into topology — not atop it.
- Cognition (#06, #19) operationalizes AI at the edge through structured signal processing.
- Identity & Path (#15–17) define connection authority without application semantics.
- Governance (#11, #18) scales the architecture to planetary level without centralized control.
- Operations (#14) translates the architecture into real-world urban deployment.

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4. Comparison with Conventional Network Architecture

The following table compares Velsanet against conventional Internet, 5G, and SDN architectures across ten architectural dimensions. This comparison is intended to assist SG13 members in identifying the points of structural departure and the corresponding study areas.

Dimension	Conventional / 5G / SDN	Velsanet
Network Model	Address-centric (IP-based)	**Topology-first (pre-coordinate)**
Data Transport	Packet switching + routing	**Physically isolated parallel optical cores**
E2E Path	Logical, shared, routed	**Physical, stateful, deterministic**
AI Integration	External overlay / service layer	**Native to structural topology (3 layers)**
Security	Upper-layer cryptography	**Physical-layer structural trust + manufacturing identity**
Governance	Centralized or federated policy	**Projection-based sovereign hierarchy (Q2–Q7)**
Parallelism	Logical multiplexing	**Physical multi-core isolation (8 channels/core)**
Failure Model	Global propagation risk	**Structural locality constraint**
Scalability	Additive logical complexity	**Structural bounded expansion**
Cognition	External AI model execution	**Network-native Cognition Cube pipeline**

4.1 Key Architectural Differentiator

The most fundamental departure is not any single feature but the ordering of design decisions. In conventional architecture, the sequence is:

Address → Topology → Connection → Intelligence (external)

In Velsanet, the sequence is:

Structure → Address → Connection → Intelligence (native)

This reversal means that security, determinism, AI embedding, and governance are structural properties — not software layers added after the fact. This is the claim that requires formal study by SG13.

5. Completeness Assessment

The following table assesses the completeness and standardization readiness of each major architectural component relative to ITU-T study item requirements.

Component	Status	Coverage	Notes for SG13
Topology-First Architecture	**Complete**	#01, #05, #07, #12	Full formal definition with mathematical proof. Ready for SG13 submission.
Physical Layer (MOCT)	**Complete**	#02, #03, #04	Hardware specification sufficient for study item proposal.
Three-Layer AI (PAI/AAI/AsAI)	**Complete**	#08, #09, #10	Conceptual and operational definitions established.
Cognition Pipeline	**Complete**	#06, #19	MVP API demonstrated at velsanet-whitepapers.onrender.com
Identity & Path Model	**Complete**	#15, #16, #17	Manufacturing-time to termination lifecycle fully specified.
Governance (IVGF/Q2–Q7)	**Complete**	#11, #18	Sovereignty-preserving projection model defined.
City-Scale Operations	**Complete**	#14	Urban deployment architecture specified.
Device Access Specs	**Planned**	#08 (noted)	Pending hardware integration milestone. Flagged in WP #08.
Prototype / Testbed	**Partial**	#19 (MVP)	Cognition Cube API exists. Full network prototype TBD.
Interoperability with 5G/6G RAN	**Pending**	—	Convergence with existing RAN standards not yet addressed. Opportunity for SG13 collaboration.

5.1 Completeness Summary

Of the ten assessed components, seven are fully complete and ready for SG13 study item consideration. Two components are at partial or planned stages. Notably, the partial status of device access specifications is explicitly acknowledged within the white paper collection itself (#08), indicating a mature and honest specification process — not a gap.

The one area requiring SG13 collaboration is interoperability with existing 5G RAN standards. This is identified as an opportunity for joint study rather than a weakness in the Velsanet specification.

6. Proposed Work Items for ITU-T SG13

Based on the architectural analysis, the following six Work Items are proposed for SG13 consideration under the IMT-2030 study programme. Each Work Item is grounded in specific white papers and addresses a distinct architectural question with direct relevance to existing SG13 Questions.

ID	Proposed Work Item Title	White Papers	Study Scope
WI-A	**Topology-First Network Space Definition**	#05, #07, #12	Study of pre-coordinate polyhedral topology as a formal basis for next-generation network space definition, in comparison with address-centric models.
WI-B	**Packet-Free Parallel Optical Path Architecture**	#02, #03, #04	Study of physically isolated parallel optical core architecture (MOCT) as an alternative to packet-switched transport for deterministic E2E connectivity.
WI-C	**Structurally Embedded AI: Three-Layer Model**	#08, #09, #10	Study of PAI/AAI/AsAI distributed AI architecture native to network topology, in contrast with AI-as-overlay approaches in current networks.
WI-D	**Network-Native Cognition Pipeline**	#06, #19	Study of SEU-based multi-channel signal decomposition and Cube synthesis as a structural cognition layer for 6G edge intelligence.
WI-E	**Manufacturing-Time Identity and Path Authority**	#15, #16, #17	Study of topology-bound identity provisioning and stateful path formation as an alternative to session-based or credential-based connection models.
WI-F	**Projection-Based Sovereign Governance (Q2–Q7)**	#11, #18	Study of structural projection hierarchy as a model for planetary-scale network governance without centralized control, preserving national sovereignty.

6.1 Priority Recommendation

For initial SG13 engagement, WI-A (Topology-First Architecture) and WI-C (Three-Layer AI) are recommended as the highest-priority study items. These two work items contain the core claims that differentiate Velsanet from all existing 6G proposals and would benefit most from formal ITU-T study and member state review.

WI-F (Sovereign Governance) is recommended for parallel engagement with ITU-T SG2 and regional bodies given its policy and sovereignty implications.

7. Recommendations for Compendium Update

7.1 Current State of the Compendium

The Velsanet Architecture Compendium (v260228) currently includes all 19 white papers in a single document. This is structurally correct and complete. However, for ITU-T SG13 readership, the following updates are recommended to maximize impact and readability:

7.2 Recommended Updates

- Add an Executive Summary (2–3 pages) at the front of the Compendium targeting SG13 members unfamiliar with Velsanet. The summary should lead with the three architectural departures (Topology-First, Packet-Free, AI-Native) and map each to a specific ITU-T Question.
- Add a White Paper Dependency Map (visual) showing how the 19 papers relate to each other. SG13 reviewers need to understand the reading order and dependencies without reading all 19 papers first.
- Add an ITU-T Cross-Reference Table mapping each white paper to the relevant SG13 Questions and existing Recommendations (Y.3001, Y.3172, M.2160, etc.).
- Add a Terminology Glossary for Velsanet-specific terms (PAI, AAI, AsAI, MOCT, Cube, SEU, DIKWEI, Q2–Q7) to reduce comprehension barriers for new readers.
- Version-stamp the Compendium as v2.0 upon incorporating the above. The current version (v260228) should be retained as the baseline reference.

7.3 What Does NOT Need to Change

The technical content of all 19 white papers is complete and does not require substantive revision. The structural logic, mathematical foundations, and architectural definitions are internally consistent. Updates are presentation and navigation enhancements only — not corrections.

8. Conclusion

The Velsanet Architecture White Paper Collection represents a complete, formally grounded, and internally consistent specification of a new class of network architecture. It is not an incremental proposal — it is a structural alternative to the foundational assumptions of the Internet.

The 19 white papers, organized across nine architectural layers, address every major dimension of network architecture from physical substrate to planetary governance. Seven of ten assessed components are fully ready for ITU-T study item consideration.

The architecture is relevant to IMT-2030 not as a competing 6G proposal but as an architectural reference — a demonstration that the constraints currently accepted as inherent to networks (packet non-determinism, external AI, address-centric identity, centralized governance) are not inherent at all. They are artifacts of a 50-year-old design choice that can be replaced.

> ***The architectural decisions made in 2026–2028 will define global connectivity for the next 30 years. Velsanet requests that Study Group 13 open a formal study track to evaluate whether the foundational assumptions of next-generation networks should be reconsidered — before those assumptions are standardized into 6G.***

Appendix A: Document Reference Index

All white papers are authored by the Velsanet Architecture Group, Republic of Korea, and are available at:

GitHub: github.com/joa337/velsanet-whitepapers

Cognition Cube MVP API: https://velsanet-whitepapers.onrender.com/docs

#01 **Velsanet Network White Paper** [v2.0] — Foundation
#02 **Wired–Wireless Convergence** [v1.0] — Physical Layer
#03 **Multi-Optical-Core Transceiver (MOCT)** [v1.1] — Physical Layer
#04 **Matrix Architecture** [v1.1] — Physical Layer
#05 **Spatial Genesis** [v1.0] — Topology
#06 **Multimodal-to-Cube Memory Architecture** [v1.0] — Cognition
#07 **Mathematical White Paper** [v1.1] — Mathematics
#08 **AI White Paper** [v1.3] — AI Architecture
#09 **Three-Layer AI System** [v1.0] — AI Architecture
#10 **Network AI White Paper** [v1.3] — AI Architecture
#11 **IVGF Conceptual Existence** [v1.0] — Governance
#12 **Foundation White Paper** [v1.1] — Foundation
#13 **Node Color White Paper** [v1.0] — Topology
#14 **City-level E2E Management Center** [v1.0] — Operations
#15 **Identity White Paper** [v1.0] — Security/Identity
#16 **Path Formation White Paper** [v1.0] — Path Management
#17 **Path Lifecycle White Paper** [v1.0] — Path Management
#18 **Network AI & Global Governance** [v1.0] — Governance
#19 **Cognition Cube System** [v1.0] — Cognition

Appendix B: ITU-T Standards Referenced

ITU-T Y.3001 (2011) Future Networks: Objectives and Design Goals
ITU-T Y.3172 (2019) Architectural framework for machine learning in future networks
ITU-T M.2160 (2023) Framework and Overall Objectives of IMT-2030
ITU-T Y.3100 Terms and definitions for IMT-2020 network
ITU-T Y.3101 Requirements of IMT-2020 network